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# Intermittent administration of parathyroid hormone (1-37) improves growth and bone mineral density in uremic rats

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**Intermittent administration of parathyroid hormone (1-37) improves growth and bone mineral density in uremic rats.**

*Background.* Parathyroid hormone (PTH) is secreted in a pulsatile fashion. Continuous infusion of PTH(1-84) resulted in a net decrease in trabecular bone volume. Differential effects have been reported following an intermittent application of PTH. We investigated the effects of a continuous infusion and of an intermittent (2 times daily subcutaneously) administration of PTH(1-37) on growth and bone mineral density (BMD) in healthy and uremic rats.

*Methods.* Two-stage subtotal nephrectomy was performed on 130 g female Sprague-Dawley rats. PTH(1-37) or solvent was administered through minipumps in sham-operated and uremic rats (60 μg/kg × day for 2 weeks). The effect of intermittent administration was tested with a subcutaneous injection of solvent: 30 μg/kg PTH(1-37) two times per day, 100 pmol calcitriol (C)/kg two times per day, or both. The length (snout-tailtip) and BMD were measured at the start of uremia and at sacrifice. BMD was measured by peripheral quantitative computer tomography at the proximal tibia, 6 and 12 mm distal of the kneejoint space. Femur bone morphology was assessed by x-rays, and calcium content was measured by atomic absorption spectrophotometry.

*Results.* Length gain was not altered by the continuous infusion of PTH. In contrast, it was significantly increased by intermittent PTH (control solvent 5.35 ± 0.37 cm vs. control PTH 6.19 ± 0.47 cm; uremia solvent 4.78 ± 0.20 cm vs. uremia PTH 6.17 ± 0.36 cm; $P < 0.05$). Intermittent PTH but not C increased BMD in uremic rats (Δ total BMD 134 + 13.3 vs. 76.3 ± 11.5 mg/mL; $P < 0.05$). X-rays revealed increased bone mass following treatment with PTH but not with C. Uremia decreased bone calcium content (64 ± 0.3 vs. 73.3 ± 2.5 mg/mL), which was normalized by PTH (80 ± 3.6 mg/mL, $P < 0.05$) but not by C (69 ± 1.9 mg/mL).

*Conclusion.* Pulsatile administration of PTH does not adversely affect, but improves longitudinal growth independent of concomitant treatment with C. At the same time PTH increases BMD and the calcium content of bone.

**Key words:** parathyroid hormone, bone mineral density, growth, uremia, trabecular bone volume

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In healthy subjects and in patients with secondary hyperparathyroidism, PTH is secreted in a pulsatile fashion [1–3]. In many hormonal systems, the pulsatile pattern of secretion has a major impact on the action of the hormone [4–6]. Likewise, the influence of exogenous PTH on bone morphology seems to be fundamentally dependent on the mode of administration. While continuous administration induced an increase in bone turnover with predominating bone resorption and a net decrease in the trabecular bone volume [7, 8], intermittent administration of PTH resulted in a net stimulation of bone formation. In normal as well as osteopenic rats, intermittent administration of PTH improved bone mass, bone mineralization, and biomechanical properties [9–13]. In clinical trials in women with postmenopausal osteoporosis, PTH as a daily injection resulted in an increase in bone mass [14].

Although severe secondary hyperparathyroidism leads to destruction of metaphyseal growth zones and by this impairs skeletal growth [15], PTH may stimulate skeletal growth in different settings. PTH stimulated DNA synthesis, cell proliferation, and colony formation in chondrocyte monolayer or suspension cultures in a biphasic manner [16]. In vivo treatment of young healthy rats with 50 μg/kg/day PTH(1-34) induced a modest increase in femora length gain after 15 days [17]. A prolonged treatment with PTH(1-34) for six months in healthy and ovariectomized rats was also reported to induce an increase in bone quality [18].

We reasoned that a continuous administration of PTH by a minipump system may not replicate the pulsatile mode of endogenous secretion. Therefore, we tested to what extent a continuous or an intermittent subcutaneous treatment with 30 μg PTH(1-37)/kg two times per day with or without the coadministration of $1{,}25(OH)_2D_3$ stimulates longitudinal growth and increases bone mineral density (BMD) in young uremic rats and control animals.

## METHODS

### Experimental animals

Female Sprague-Dawley rats (Fa. Ivanovas, Kissleg/Allgäu, Germany) were used for the experiments. They were kept individually in a light-controlled (12-h on/off cycles), temperature-controlled (24°C), and humidity-controlled (70%) room and were allowed to adapt to the new environment for seven days.

All animals had free access to food (Altromin C 1000 diet; Altromin Co., Lage, Lippe, Germany) and deionized water, except for the control animals in experiment 2. The diet contained 13800 kJ/kg, 17% protein (wt/wt), 0.95% calcium, 0.75% phosphorus, and 500 IU/kg vitamin $D_3$. The day prior to the first operation, the animals were assigned to the different experimental groups according to their body weight. A two-stage subtotal nephrectomy (NX) was performed as described earlier [19]. Seven days after subtotal NX, the contralateral kidney was removed. The mean weight of the animals was 133.5 ± 6.4, 130 ± 8.7, 134 ± 9.2, and 125 ± 7.9 g in experiments 1, 2, 3, and 4 when uremia was established. Control animals were sham operated (renal decapsulation).

### Experiment 1

*Effect of continuous parathyroid hormone in uremic and healthy animals.* In healthy and uremic animals, osmotic minipumps (Alzet 2002; ALZA Corporation, Palo Alto, CA, USA) were implanted subcutaneously at the day of the second operation for subtotal NX. The minipumps either infused 60 μg/kg × day of human recombinant PTH(1-37) (Boehringer Co., Mannheim, Germany) at a pumping rate of 0.5 μL/h or solvent ($N = 7$ per group). The duration of the experiment was 14 days.

### Experiment 2

*Effect of intermittent parathyroid hormone in uremic and healthy animals.* Uremic and sham-operated rats were randomized for treatment with twice daily vehicle solution subcutaneously or with PTH [60 μg PTH(1-37)/kg × day given in 2 doses subcutaneously; $N = 7$ per group]. The duration of experiment (uremia and treatment) was 14 days. The sham-operated control animals were pair fed.

### Experiment 3

*Uremic animals concomitantly treated with parathyroid hormone and 1,25(OH)$_2$D$_3$.* All animals underwent subtotal NX and were randomized into four different treatment groups ($N = 11$ per group): solvent, 200 pmol 1,25(OH)$_2$D$_3$ kg body wt/day subcutaneously divided in two doses (Roche Company, Grenzach-Wyhlen, Germany); 60 μg PTH(1-37)/kg body wt/day subcutaneously divided in two doses; and the combination of both, PTH(1-37) and 1,25(OH)$_2$D$_3$. The duration of experiment was 14 days.

### Experiment 4

*Effect of parathyroid hormone and/or 1,25(OH)$_2$D$_3$ on bone mineral density in uremic rats.* Subtotal nephrectomized animals were randomized into four treatment groups on the analogy of experiment 3 ($N = 11$ per group): solvent, 1,25(OH)$_2$D$_3$, PTH, 1,25(OH)$_2$D$_3$ + PTH. A control group with 11 sham-operated solvent-treated animals was added. The duration of the experiment was three weeks. BMD was measured by peripheral quantitative computer tomography (pQCT; discussed later in this article) at start of uremia and prior to sacrifice.

### Measurements

All measurements of body weight were performed in nonfasted animals in the morning. The length gain was measured under general anesthesia (30 mg/kg Ketanest; Park Davis Company, Berlin, Germany; and 2 mg/kg Valium; Hoffmann la Roche, Grenzach-Wyhlen, Germany) during the second-stage operation and prior to sacrifice. The last hormone injection was performed 12 hours before sacrifice. Blood from the abdominal aorta was collected for serum analyses of calcium, phosphorous, sodium, potassium, alkaline phosphatase, serum creatinine, and serum urea. During the last 24 hours of experiment 3, the animals were kept in metabolic cages to obtain 24-hour urine samples. Serum and urine biochemistry were analyzed using a multichannel autoanalyzer (Synchron CX7; Beckmann, Hamburg, Germany). Creatinine was determined by a kinetic method according to Jaffé without deproteinization; the within-assay coefficient of variation was <3%. PTH was measured using an aminoterminal sensitive immunoradiometric assay with one receptor specific for the rat aminoterminal region 1 through 13 and a second antibody specific for the midregional sequence 14 through 34 (Nichols Institute, San Juan Capistrano, CA, USA). This assay did not discriminate between endogenous and exogenous PTH(1-37). The intra-assay coefficient of variation was 5.3%.

### Evaluation of bone for mineral density and calcium content

Bone architecture was evaluated by x-rays of the right femur after sacrifice, and femur volume was measured by water immersion. The specimen were dried at 80°C for 24 hours, weighed, and then ashed in a muffle (Thermicon T, Heraeus) at 600°C for 24 hours. Calcium content was determined using an atomic absorption spectrophotometer (2100 AAS; Perkin-Elmer, Norwalk, CT, USA), and bone calcium content was defined as a percentage of ratio of bone calcium to volume.

Bone mineral density was evaluated by peripheral pQCT. Measurements were carried out in the proximal tibia metaphysis (6 mm distal to the kneejoint space)

**Table 1.** Effect of treatment with parathyroid hormone (1-37) [PTH(1-37; 30 μg/kg b.i.d. s.c.)] for 14 days on growth and serum biochemistry in sham-operated pair fed controls and uremic rats (experiment 2, $N = 7$ per group)

| | Controls solvent | Controls PTH | Uremia solvent | Uremia PTH |
|---|---|---|---|---|
| Serum creatinine *mg/dL* | $0.38 \pm 0.02^a$ | $0.45 \pm 0.03^a$ | $0.72 \pm 0.02^b$ | $0.67 \pm 0.05^b$ |
| Serum calcium *mmol/L* | $2.47 \pm 0.07^a$ | $2.51 \pm 0.08^a$ | $2.44 \pm 0.17^a$ | $2.26 \pm 0.15^a$ |
| Serum phosphate *mmol/L* | $2.44 \pm 0.09^a$ | $2.74 \pm 0.19^a$ | $2.66 \pm 0.14^a$ | $2.48 \pm 0.17^a$ |
| Δ Weight gain *g* | $44.4 \pm 2.5^a$ | $52.8 \pm 8.1^a$ | $41.2 \pm 3.2^a$ | $51.3 \pm 7.9^a$ |
| Food conversion ratio *g/g* | $0.23 \pm 0.01^a$ | $0.26 \pm 0.03^a$ | $0.23 \pm 0.01^a$ | $0.26 \pm 0.02^a$ |
| Length gain *cm* | $5.35 \pm 0.37^a$ | $6.19 \pm 0.47^b$ | $4.78 \pm 0.20^a$ | $6.17 \pm 0.36^b$ |

Data given as mean ± SEM. The overall effects of the different treatment options were analyzed by ANOVA, followed by Duncan's test for multiple comparisons to analyze the differences between each treatment group. Superscript letters indicate significant differences between groups. Values sharing common superscripts are not significantly different, while values without common superscripts are significantly different ($P < 0.05$).

and diaphysis (12 mm distal to the kneejoint) 24 hours prior to the second-stage operation and at sacrifice using an XCT-960A from Stratec (Pforzheim, Germany). A voxel size of 0.092 × 0.092 × 1 mm and threshold for cancellous bone of 0.700 were chosen throughout the experiment. The threshold for cortical bone was 0.930 with the exception of 0.750 at the 6 mm site at the endosteal layer in order to delineate cancellous from cortical bone (Contour mode 1, Peel mode 2, Cortical mode 2). The coefficient of variation with repeated positioning of the limbs was found to be 2.1% for total density, 3.5% for trabecular density, and 1.8% for cortical density.

Since it has not been possible to perform pQCT in all animals at the same day, the experiment was performed within three days. This means that the subtotal NX, pQCT measurements, and sacrifice were performed sequentially and in a randomized fashion for all treatment groups.

### Statistics

Data are given as mean ± SEM if not indicated otherwise. The overall effects of the different treatment options were analyzed by analysis of variance, followed by Duncan's test for multiple comparisons to analyze the differences between each treatment group. $P < 0.05$ was accepted as statistically significant.

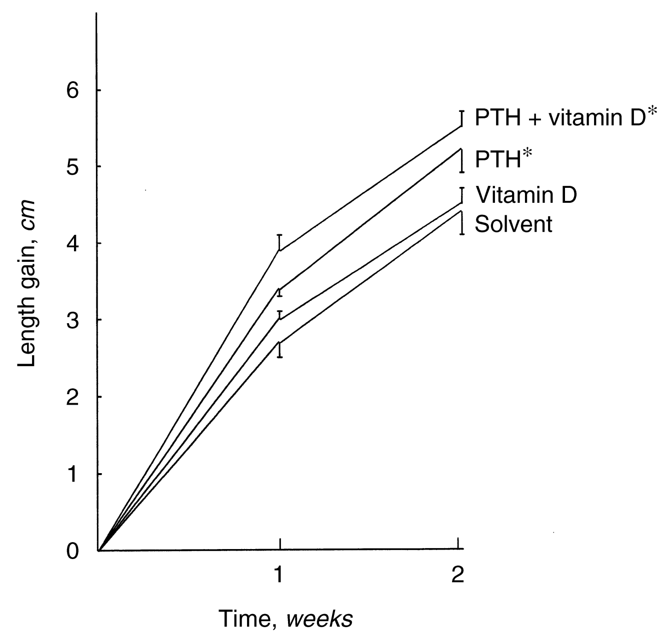


**Fig. 1. Length gain in uremic rats treated either with solvent, PTH(1-37) subcutaneously (30 μg/kg b.i.d. subcutaneously), 1,25(OH)$_2$D$_3$ (100 pmol/kg b.i.d. subcutaneously), or both (experiment 3, $N = 11$ per group).** Data are given as mean ± SEM. The overall effects of the different treatment options were analyzed by analysis of variance, followed by Duncan's test for multiple comparisons to analyze the differences between each treatment group. The asterisk denotes significant difference versus solvent and 1,25(OH)$_2$D$_3$-treated animals ($P < 0.05$).

## RESULTS

### Effect of continuous parathyroid hormone (1-37) on growth in healthy and uremic animals (experiment 1)

In healthy and uremic ad libitum fed animals, 60 μg/kg × day PTH(1-37) or solvent was constantly infused via minipumps. After two weeks of PTH(1-37) treatment, no significant differences were noted in mean weight gain (data not given) and length gain [control animals: PTH 4.2 cm (2.7 to 4.5), solvent 3.7 cm (3.2 to 4.4); uremic animals: PTH 3.9 cm (3.0 to 5.1), solvent 3.5 cm (3.3 to 4.3), median/range, $P$ = NS]. PTH plasma levels at the end of the experiment were 143 ± 26 pg/mL in control animals with solvent infusion compared with 385 ± 119 pg/mL in animals with the PTH(1-37) infusion. In solvent-treated uremic rats, mean PTH plasma levels were 240 ± 41 pg/mL compared with 526 ± 130 pg/mL in animals with continuous PTH(1-37) infusion ($P < 0.05$).

### Effect of intermittent parathyroid hormone (1-37) on growth in uremic animals and sham-operated pair fed controls (experiment 2)

To evaluate whether intermittent PTH(1-37) administration has a positive effect on growth, PTH(1-37) was given subcutaneously (2 × 30 μg/kg × day) to uremic and sham-operated, pair-fed control animals. The length gain of those animals was compared with solvent-treated controls (Table 1).

**Table 2.** Serum and urine biochemistry, body weight gain, and food conversion ratio of uremic rats treated either with solvent, PTH(1-37) (30 μg/kg b.i.d. s.c.), 1,25 vitamin $D_3$ (100 pmol/kg b.i.d. s.c.), or both (experiment 3, $N = 11$ per group)

| | Solvent | Vitamin $D_3$ | PTH | PTH + vitamin $D_3$ |
|---|---|---|---|---|
| Weight gain *g* | $37.4 \pm 1.9^{a}$ | $40 \pm 1.8^{ab}$ | $45.2 \pm 2.9^{b}$ | $44.5 \pm 6.1^{b}$ |
| Food intake *g* | $213 \pm 6^{a}$ | $201 \pm 6^{a}$ | $210 \pm 7^{a}$ | $214 \pm 5^{a}$ |
| Food conversion ratio *g/g* | $0.18 \pm 0.01^{a}$ | $0.19 \pm 0.01^{ab}$ | $0.21 \pm 0.01^{b}$ | $0.21 \pm 0.01^{b}$ |
| Serum creatinine *mg/dL* | $0.55 \pm 0.03^{a}$ | $0.51 \pm 0.02^{a}$ | $0.49 \pm 0.03^{a}$ | $0.53 \pm 0.01^{a}$ |
| Serum urea *mg/dL* | $70.8 \pm 3.4^{a}$ | $63 \pm 2.9^{ab}$ | $57.4 \pm 3.7^{b}$ | $57.5 \pm 2.1^{b}$ |
| Serum calcium *mmol/L* | $2.59 \pm 0.04^{a}$ | $2.52 \pm 0.05^{a}$ | $2.44 \pm 0.05^{a}$ | $2.47 \pm 0.03^{a}$ |
| Serum phosphate *mmol/L* | $2.39 \pm 0.06^{a}$ | $2.4 \pm 0.03^{a}$ | $2.39 \pm 0.04^{a}$ | $2.64 \pm 0.07^{b}$ |
| Serum AP *U/L* | $263 \pm 24^{a}$ | $243 \pm 9^{a}$ | $275 \pm 20^{a}$ | $292 \pm 37^{a}$ |
| Urine creatinine *mg/kg/day* | $0.95 \pm 0.05^{a}$ | $1.02 \pm 0.05^{a}$ | $1.05 \pm 0.07^{a}$ | $1.05 \pm 0.09^{a}$ |
| Urine urea *mg/kg/day* | $63.7 \pm 4.4^{a}$ | $63.5 \pm 3.4^{a}$ | $51.5 \pm 9.9^{a}$ | $66.1 \pm 4.6^{a}$ |
| Urine calcium *μmol/kg/day* | $1.34 \pm 0.38^{a}$ | $3.65 \pm 1.45^{a}$ | $3.10 \pm 0.85^{a}$ | $3.00 \pm 0.47^{a}$ |
| Urine phosphate *μmol/kg/day* | $118 \pm 6^{a}$ | $118 \pm 15^{a}$ | $160 \pm 23^{a}$ | $138 \pm 8^{a}$ |

Data are given as mean ± SEM. The overall effects of the different treatment options were analyzed by ANOVA, followed by Duncan's test for multiple comparisons to analyze the differences between each treatment group. Superscript letters indicate significant differences between groups. Values sharing common superscripts are not significantly different, while values without common superscripts are significantly different ($P < 0.05$).

Parathyroid hormone (1-37) did not have any systematic effect on serum calcium and serum phosphate. It did not influence body weight gain and food conversion ratio. However, there was a significant increase in length gain in uremic animals, as well as in sham-operated pair fed controls. PTH plasma levels six hours after a subcutaneous injection were 407 ± 208 pg/mL. At sacrifice, which was about 12 hours after the last injection, iPTH levels were 290 ± 87 pg/mL.

### Effect of concomitant treatment with 1,25(OH)$_2$D$_3$ and parathyroid hormone (1-37) on growth in uremic animals (experiment 3)

As in experiment 2, intermittent subcutaneous PTH(1-37) significantly increased length gain in uremic animals (Fig. 1), whereas 1,25(OH)$_2$D$_3$ did not. When PTH(1-37) was given together with 1,25(OH)$_2$D$_3$, the effect on length gain was slightly but not significantly better compared with PTH(1-37) given alone.

In this experiment, PTH(1-37) had a slightly positive effect on the ratio of weight gain per food intake and cumulative weight gain. There was no significant effect of either 1,25(OH)$_2$D$_3$ or PTH(1-37) on serum biochemistry, as well as on biochemical findings of the 24-hour urine (Table 2).

### Effect of 1,25(OH)$_2$D$_3$ and parathyroid hormone (1-37) on bone mineral density and bone calcium content (experiment 4)

The bone architecture of the right femora was evaluated by postmortem x-rays of the animals. As seen from Figure 2, 1,25(OH)$_2$D$_3$ did not have any obvious effect on bone structure and bone x-ray density. In contrast, there was a striking increase of cortical thickness as well as on metaphyseal x-ray density in animals treated with PTH(1-37) either alone or in combination with 1,25(OH)$_2$D$_3$.

Peripheral quantitative computer tomography measurements were carried out in the proximal tibia metaphysis (6 and 12 mm distal to the kneejoint space) at the start and end of the experiment. Whereas there was no significant difference for BMD at the start, animals treated with PTH(1-37) had a significantly higher BMD at the end of the experiment. Figure 3 and Table 3 give the increase in BMD within the three weeks of the experiment.

The increase in total BMD was the consequence of a combined increase in cortical density and trabecular density (Fig. 3 and Table 3). The increase was noted at both measurement points, but was more expressed in the metaphysis adjacent to the growth plate. Furthermore, the intermittent treatment with PTH(1-37) induced a significant increase in the periosteal circumference and a significant decrease in the endosteal circumference, both leading to a substantial increase in the cortical area. In contrast, 1,25(OH)$_2$D$_3$ given alone or in combination with PTH did not have a positive effect on BMD and the cortical area.

In accordance with the changes in BMD, the dry weight, as well as the calcium content of bone, was significantly increased with PTH(1-37) but not with 1,25(OH)$_2$D$_3$ (Table 3).

## DISCUSSION

The present experiments demonstrate that intermittent but not continuous administration of PTH(1-37) increases growth, bone calcium content, and BMD in uremic animals. Concomitant treatment with 1,25(OH)$_2$D$_3$ did not influence the effects of intermittent PTH.

Parathyroid hormone is a major humoral regulator of calcium homeostasis. In addition, it has an anabolic action that has been known for decades, but has been forgotten for awhile.

Bauer, Aub, and Albright first demonstrated in 1929

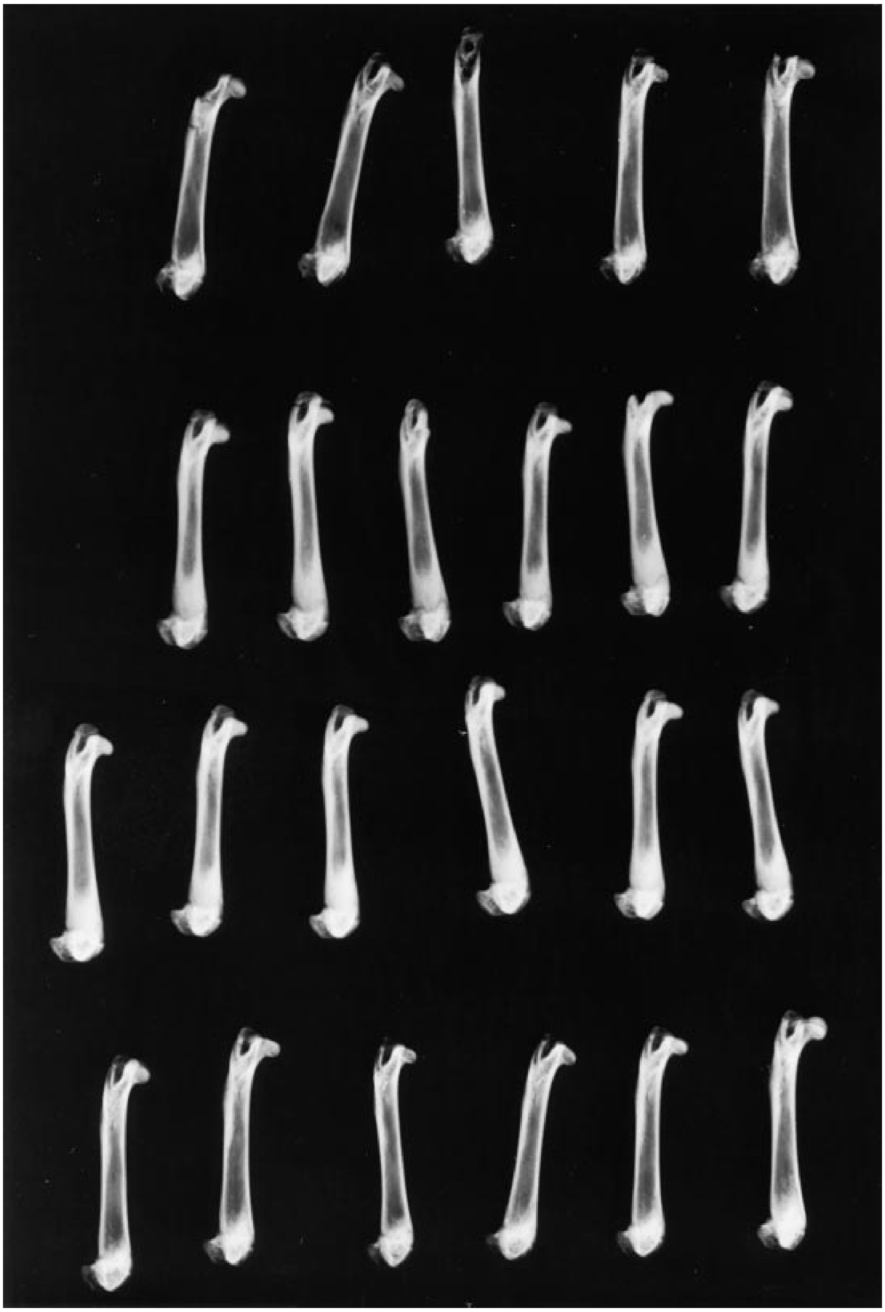

**Fig. 2. X-ray analysis of the femora from uremic animals treated with solvent (upper row) PTH(1-37) (30 μg/kg b.i.d. subcutaneously, second row), 1,25(OH)$_2$D$_3$ (100 pmol/kg b.i.d. subcutaneously, bottom row), or both (third row, experiment 4).** PTH(1-37) markedly increased bone density and cortical bone thickness.

that PTH could increase skeletal calcium by injecting parathyroid extract into rats [20]. This result has been confirmed 1932 by Selye, who extended the experiments to dogs [21]. The anabolic effects of PTH has engendered interest in its pharmacological use during the last decade after recombinant human PTH has become available.

The biological effect of exogenously administered PTH can be anabolic or catabolic, depending on the mode of administration. Animal studies have convincingly shown that intermittent PTH administration increases bone mass, while continuous exposure had a marked catabolic activity on the skeleton [7–13, 17]. In our experiments, intermittent PTH had an anabolic effect on the skeleton, but growth was not negatively affected by continuous PTH. This might be due to the fact that the experiment lasted for only two weeks and that plasma PTH levels were only increased twofold to threefold by the amount of PTH infused by minipumps per day. Although complete plasma concentration profiles were not measured following intermittent subcutaneous injections, the 6-hour and 12-hour measurements showed a transient increase in plasma concentration of PTH.

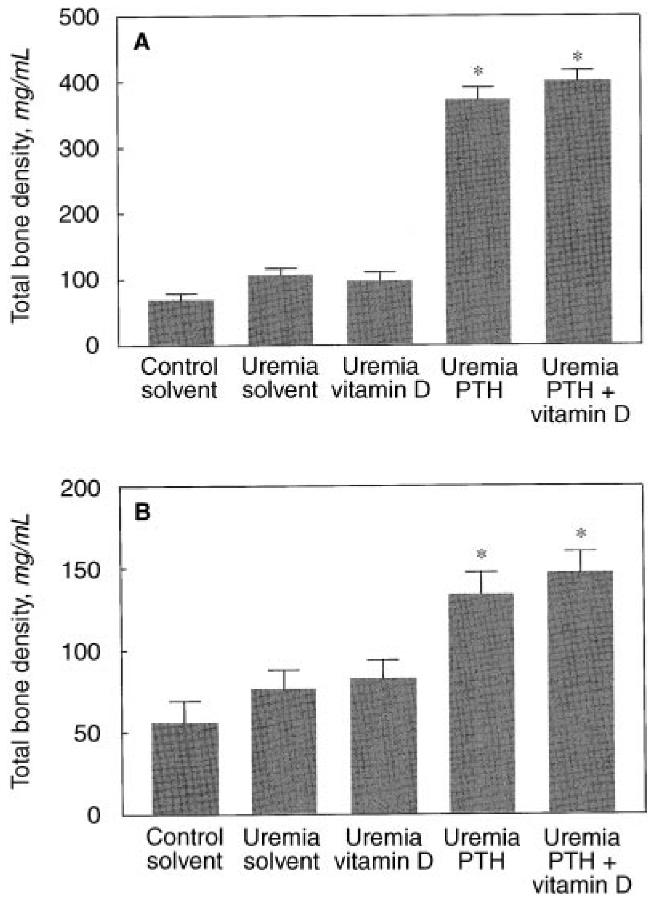


**Fig. 3. Increase in total bone density per volume obtained by pQCT-measurements at 6 (upper panel) and 12 mm (lower panel) distal to the knee joint space (experiment 4).** The overall effects of the different treatment options were analyzed by analysis of variance, followed by Duncan's test for multiple comparisons to analyze the differences between each treatment group. The asterisk denotes significant difference vs. solvent and $1,25(OH)_2D_3$ animals ($P < 0.05$). The absolute increase of bone mineral density was significant in all groups. However, the increase was significantly higher in PTH(1-37)-treated animals.

The mechanism by which the profound anabolic effect of intermittent in contrast to continuous PTH administration is achieved remains to be elucidated. PTH is secreted in a tonic and pulsatile fashion [2]. The oscillatory pattern of PTH secretion is profoundly disturbed in uremia [3]. Moreover, a bimodal diurnal rhythm of PTH secretion exists with maximal plasma concentrations in the late afternoon and in the earlier morning [22]. In principle, the pattern-dependent metabolic action of PTH may be determined by an absolute peak concentration, the nadir concentration, the amplitude, the length of the time periods for which certain threshold peak or nadir concentration ranges are transgressed, or by the dynamics, that is, the rate of change in plasma PTH concentrations. Of note, the nadir levels have been shown to determine the pattern-specific effects of pulsatile versus continuous growth hormone on specific hepatic enzyme induction in rats [23]. Another possible explanation of the observed effects of intermittent PTH(1-37) administration would be that the transient suppression of endogenous intact PTH secretion, rather than actions of the exogenously administered hormone, was responsible for the changes observed. We previously observed rapid down-regulation of endogenous iPTH secretion after administration of PTH(1-37) in the rat (unpublished results), and such autoinhibitory effects have also been reported in clinical studies [24]. Although we cannot rule out an indirect action of PTH(1-37) via induction of fluctuations of endogenous PTH, this mechanism appears to be less likely since PTH(1-37) contains the amino-terminal fragment and the midregional fragment, which are responsible for the metabolic actions of PTH in growth cartilage and bone [16, 25]. Moreover, since PTH(1-37) apparently down-regulates endogenous PTH secretion by binding to parathyroidal PTH autoreceptors, it is likely that the hormone receptor interaction does not differ to a major degree between PTH(1-84) and PTH(1-37). On the other hand, intermittent suppression of endogenous PTH may be a promising therapeutic alternative, especially in patients with hyperparathyroidism, which would mimic the pulsatile patterns induced by exogenous PTH administration. The recent advent of calcium receptor mimetic drugs [26], which selectively and reversibly suppress endogenous PTH secretion, opens a new perspective for studies of the differential metabolic effects of different patterns of endogenous plasma PTH concentrations.

In experiments with nonuremic rodents, intermittent exogenous PTH increased not only trabecular bone mass but also cortical bone mass [27, 28]. This might be in contrast to expectations of nephrologists who see an increase of trabecular bone volume in iliac crest biopsies in hyperparathyroid patients [29, 30] on the one hand, but periostal resorptions of the cortical layer in radiological investigations on the other hand [15, 31]. Our study confirms the experience from experimental studies and extends these to the uremic skeleton. It has been shown that the effects of PTH are dose and time dependent, leading to a major effect during the first six treatment weeks, primarily at the vertebrae and the upper tibia [32]. Peripheral QCT measurements in our experiments indicated an increase both in cortical and trabecular bone density at the proximal tibia. It is notable that the cortical bone became thicker since not only the periosteal circumference increased, but the endosteal circumference decreased (Table 3). It cannot be decided whether this apparent shrinking of the endosteal circumference was due to corticalization of the trabecular bone or was in part a measuring artifact due to the insufficient resolutions depending on the voxel size of the equipment. In any case, the increased total BMD was paralleled by an increase in the calcium content of the bone, as verified

**Table 3.** Measurement of bone calcium content and of changes (Δ) in BMD within 3 weeks in solvent treated sham-operated rats and uremic rats either treated with solvent, PTH (1-37) s.c. (30 μg/kg b.i.d. s.c.), 1,25 vitamin $D_3$ (100 pmol/kg b.i.d. s.c.) or both for 3 weeks (experiment 4, $N = 11$ per group)

| | Control solvent | Uremia solvent | Uremia $1,25(OH)_2D_3$ | Uremia PTH | Uremia PTH + $1,25(OH)_2D_3$ |
|---|---|---|---|---|---|
| Bone dry weight/volume *mg/mL* | $877 \pm 20^a$ | $771 \pm 24^b$ | $921 \pm 38^a$ | $899 \pm 24^a$ | $944 \pm 21^a$ |
| Bone calcium content *mg/mL* | $73.3 \pm 2.5^{ab}$ | $64 \pm 0.3^b$ | $69 \pm 1.9^b$ | $80 \pm 3.6^a$ | $82 \pm 2.6^a$ |
| PQCT measurements at 6 mm | | | | | |
| Δ Total bone content per slice *mg/mm* | $2.26 \pm 0.16^a$ | $2.02 \pm 0.13^a$ | $1.96 \pm 0.14^a$ | $6.32 \pm 0.32^b$ | $6.91 \pm 0.45^b$ |
| Δ Total bone area *mm*$^2$ | $3.26 \pm 0.15^a$ | $2.09 \pm 0.28^b$ | $2.13 \pm 0.16^b$ | $3.92 \pm 0.28^{ac}$ | $4.12 \pm 0.39^c$ |
| Δ Periosteal circumference *mm* | $1.74 \pm 0.09^a$ | $1.17 \pm 0.16^b$ | $1.24 \pm 0.09^b$ | $2.14 \pm 0.14^{ac}$ | $2.21 \pm 0.19^c$ |
| Δ Endosteal circumference *mm* | $1.06 \pm 0.07^a$ | $0.15 \pm 0.17^a$ | $0.50 \pm 0.13^a$ | $-3.19 \pm 0.33^b$ | $-3.16 \pm 0.53^b$ |
| Δ Cortical density *mg/ccm* | $103 \pm 8^a$ | $123 \pm 10^{ab}$ | $136 \pm 9^{ab}$ | $157 \pm 15^b$ | $199 \pm 9^c$ |
| Δ Trabecular density *mg/mL* | $11.8 \pm 8.6^a$ | $-8.98 \pm 5.6^a$ | $-3.05 \pm 5.5^a$ | $134 \pm 15.0^b$ | $141 \pm 17.2^b$ |
| Δ Cortical area *mm*$^2$ | $1.84 \pm 0.11^a$ | $1.91 \pm 0.14^a$ | $1.54 \pm 0.14^a$ | $6.93 \pm 0.44^b$ | $7.11 \pm 0.53^b$ |
| PQCT measurements at 12 mm | | | | | |
| Δ Total bone content per slice *mg/mm* | $2.35 \pm 0.05^a$ | $1.82 \pm 0.07^b$ | $1.94 \pm 0.09^b$ | $2.68 \pm 0.09^c$ | $2.72 \pm 0.11^c$ |
| Δ Total bone area *mm*$^2$ | $2.98 \pm 0.08^a$ | $2.15 \pm 0.13^b$ | $2.18 \pm 0.14^b$ | $2.75 \pm 0.11^a$ | $2.71 \pm 0.17^a$ |
| Δ Periosteal circumference *mm* | $2.21 \pm 0.07^a$ | $1.63 \pm 0.10^b$ | $1.71 \pm 0.09^b$ | $2.08 \pm 0.08^a$ | $2.03 \pm 0.11^a$ |
| Δ Endosteal circumference *mm* | $1.35 \pm 0.08^a$ | $0.88 \pm 0.13^b$ | $0.83 \pm 0.11^b$ | $0.74 \pm 0.12^b$ | $0.69 \pm 0.13^b$ |
| Δ Cortical area *mm*$^2$ | $1.77 \pm 0.07^a$ | $1.38 \pm 0.07^b$ | $1.49 \pm 0.07^b$ | $2.14 \pm 0.08^c$ | $2.11 \pm 0.07^c$ |

Data are given as mean ± SEM. The overall effects of the different treatment options were analyzed by ANOVA, followed by Duncan's test for multiple comparisons to analyze the differences between each treatment group. Superscript letters indicate significant differences between groups. Values sharing common superscripts are not significantly different ($P < 0.05$).

by atomic absorption measurements. Finally, the radiological work-up provides evidence that the major changes in BMD were located within the metaphyseal areas and within the cortical bone.

Our measurements of BMD do not allow any information on bone quality. We also do not have a micromorphometric analysis of the bone. Nevertheless, it is of interest to note that Capozza et al reported not only on an increase in BMD, but also showed an increased mechanical strength of the femur in immobilized and overloaded femurs of PTH(1-38) treated rats [33].

It is more difficult to interpret clinical studies on PTH treatment. In a prospective study in 21 osteoporotic women treated for 6 to 24 months with a once daily injection of PTH(1-34), there was a significant increase in trabecular bone volume, as assessed by paired biopsies, and a striking increase in the rate of new bone formation, as measured by calcium kinetics [34]. In 1997, Lindsay et al reported on the effect of 400 IU PTH(1-34) in a randomized, controlled study in 34 women with postmenopausal primary osteoporosis over a three-year period. They demonstrated an increment of both cortical and trabecular bone induced by PTH [35]. However, in both studies, the patients were on estrogen therapy during the treatment with PTH, and it might be possible that the positive effect on cortical bone was dependent on the concomitant estrogen therapy, whereas an isolated PTH therapy might have the risk of cortical erosions.

Because of the concern and the uncertainty regarding the net effect of PTH on calcium balance, a "combination therapy" with PTH and a second "antiresorptive drug" has been investigated frequently [10, 14, 28]. Among those studies, several attempts to combine PTH with $1,25(OH)_2D_3$ have been performed. A positive effect of the concomitant therapy has been reported in osteoporotic women [36]; however, these studies were uncontrolled. Even animal studies do not provide a definite answer about whether $1,25(OH)_2D_3$ has a direct anabolic effect on bone or whether its effect is mediated indirectly through the improvement of calcium balance. There is evidence from vitamin D-deficient rats and from vitamin D receptor knockout animals that $1,25(OH)_2D_3$ is not essential for the normal function of bone cells [37–39]. In vitamin D-deficient female rats, PTH increased bone formation and improved mineral balance obviously without any need of $1,25(OH)_2D_3$ [13].

In the present study in uremic animals, $1,25(OH)_2D_3$ did not effect BMD when given either alone or in combination with PTH(1-37), at least at the given dose. It is interesting that Gunness-Hey et al did also not find an increase in bone mass in young healthy rats with the same dose of $1,25(OH)_2D_3$ when given alone or concomitantly with PTH [40]. Higher doses increased trabecular bone dry weight and hydroxyproline content; however, mineralization was impaired, and rats became hypercalcemic. Erben, Bromm, and Stangassinger provided some evidence that the pharmacological administration of $1,25(OH)_2D_3$ was able to elicit a direct anabolic response in ovariectomized rats, whereas calcium supplementation alone was without therapeutic effect [41].

Our study is, to our knowledge, the first to demonstrate a positive effect of intermittent PTH administration on longitudinal growth of uremic animals. These observations confirm indirectly recent in vitro findings of our group where PTH dose-dependently stimulated proliferation of growth plate chondrocytes in a biphasic manner [16]. There are also clinical observations supporting a growth promoting effect of PTH. Children with hypoparathyroidism and pseudohypoparathyroidism are usually growth retarded, although this may be due in part to a polyglandular disease that is often noted in those patients [42, 43]. It is also of note that PTH plasma concentrations were positively correlated with growth velocity in dialyzed children [44]. There is no contradiction when we reported earlier that the severest forms of renal hyperparathyroidism can lead to metaphyseal destruction and growth arrest [15].

In conclusion, our study confirms and extends to the uremic skeleton of rodents that PTH not only stimulates bone resorption but also bone apposition. In addition, we found a positive effect on longitudinal growth. The mode of administration of exogenous PTH is a major determinant of its action and its potent anabolic effect, for example, improvement in growth and BMD. It remains to be shown in further studies whether the modulation of PTH levels into a more pulsatile pattern is able to improve the low bone turnover state in uremic patients as it does in postmenopausal osteoporosis. Since a cancerogenic potential of PTH(1-37) has recently been noted in rodents after long-term administration (personal communication, Werner Blum, M.D.), recombinant human PTH will not be available for clinical studies in the near future. It is, however, tempting to speculate that similar effects as presented in this study might be obtained by modulation of iPTH plasma levels with intermittent administration of calcimimetic agents [45].

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